

Professional Summary

HPC and GPU software engineer with expertise in C++ (C++20/23), CUDA/ROCm, and performance engineering for large-scale scientific applications.

Experience in developing performance-portable software across CPU/GPU architectures, combined with hands-on work on Linux-based HPC systems, including cluster design, deployment, and distributed workloads.

Background in numerical methods and computational physics, with contributions to peer-reviewed research and in-house high-performance scientific software. Experience in mentoring and technical supervision within HPC environments.

Education

- 11/2021 – 02/2025 **PhD in Mathematical Models and Methods in Engineering**, *Laboratory for Modelling and Scientific Computing (MOX)*, Mathematics Department, Politecnico di Milano
Design and development of C++20 code for GPUs and CPUs (Thrust, std::execution/TBB, CUDA, HIP).
PhD Thesis: “Material Point Method for Compressible Flows: a Portable, Massively Parallel Implementation for High-Performance Computing Architectures.”
Supervisor: prof. Carlo de Falco, carlo.defalco@polimi.it
- 2021 **National Habilitation**, Senior Nuclear Engineer
- 10/2020 – 10/2021 **MSc in Nuclear Eng.**, Nuclear System Physics, *Politecnico di Milano*, 107/110
- 09/2017 – 10/2020 **MSc in Math. Eng.**, Computational Sciences and Engineering, *Politecnico di Milano*, 110/110
- 09/2013 – 07/2017 **BSc in Ingegneria Matematica**, *Politecnico di Milano*, 110/110

Professional Experience

- 12/2023 – present
(Permanent Position) **Technical Manager - MOX Laboratory for Modelling and Scientific Computing**, Mathematics Department, Politecnico di Milano
- Development and optimisation of scientific software using modern C++ and GPU technologies (CUDA, ROCm), with a focus on performance portability
 - Design, deployment, and maintenance of Linux-based HPC systems for scientific computing, ensuring performance, reliability, and efficient resource utilisation in multi-node environments
 - HPC cluster design, deployment, and system integration (on-premise systems)
 - Configuration and management of HPC software stacks and workload managers, with OpenPBS as the primary scheduler and experience on SLURM-based systems (CINECA Leonardo Booster)
 - Collaboration with academic and industry partners on HPC and scientific computing projects
 - Principal Investigator (PI) of the “MatBa” project, enabling access to national HPC resources and supporting research computing workflows
 - Leadership of an HPC infrastructure renewal project, securing funding for modern GPU-enabled systems and high-speed interconnects
 - Supervision of technical staff for Linux system administration and deployment
 - Co-supervision of MSc and PhD students on HPC and GPU-related topics
- 2022 – present **Teaching Assistant - Scientific Computing MSc Courses**, Politecnico di Milano
- Delivered advanced C++ and numerical methods labs at MSc level — github.com/pacs-course/pacs-Labs/tree/main/Labs and github.com/HPC-Courses/AMSC-Labs/tree/main/Labs/2025-26
 - Covered advanced topics in C++23, MPI, GPU programming, build systems, debugging and performance tools, modern HPC software stacks and abstraction layers.
 - Supervision and mentoring of graduate students in advanced scientific computing and HPC topics

Key Technical Skills

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|-------------------|--|
| Programming | C++20/23, CUDA-C, HIP/ROCm, MPI, Eigen, Parallel Algorithms (Thrust, std::execution/TBB), C, Make, shell, Octave/Matlab, Python (scripting), pybind11, and Numba |
| Software & Tools | git/GitHub, Podman, Apptainer, Spack, CMake, Doxygen, PBS/SLURM, gdb, lcov, valgrind, Nsight |
| Numerical Methods | FEM, DG, PIC/MPM, Numerical Linear Algebra, Symplectic Integrators, Runge-Kutta methods |

- Systems Linux (Debian, Arch, RHEL), HPC cluster deployment, virtualisation tools (iDRAC, iLO, xClarity)
- Scientific Software
- [SOLPS-ITER](#), coupled Finite Volumes and Monte Carlo code for the simulation of boundary plasma in magnetic confinement fusion devices, including [B2.5](#) (plasma) and [EIRENE](#) (WCU MC)
 - [PyFR](#), solver for advection–diffusion using Flux Reconstruction, designed for high-performance and portability across CPU/GPU architectures via a Python-based domain-specific templating layer

Professional Development

- Red Hat & Coursera RHEL, Ansible, Kubernetes, Amazon Simple Storage Service (S3) (2025-2026)
- NVIDIA Courses Scaling Workloads Across Multiple GPUs with CUDA C++, GPU Acceleration with the C++ Standard Library, Building Container Images for HPC, Accelerated Computing with CUDA Python, Accelerating CUDA C Applications with Concurrent Streams, and additional courses (2024–2025)
- CINECA Courses Build Systems and Package Managers in HPC, Parallel Computing on Traditional Core-Based and Emerging GPU-Based Architectures, Numerical Methods for Parallel CFD (2021-2023)
- Conferences and Workshops Continuous learning through attendance at various industry workshops, EuroHPC hackathons, and scientific conferences on HPC, AI, Scientific Machine Learning, Quantum Computing

Publications and Conferences

- 2025 **Workshop Poster (selected)**, *Portable Parallel Computing: C++ Strategies for Cross-Architecture Performance*, Baioni P.J., Benacchio T., Capone L., de Falco C., [E4: AI, HPC & QUANTUM 2025](#)
- 2024 **Journal Paper**, *Portable, massively parallel implementation of a material point method for compressible flows*, Baioni P.J., Benacchio T., Capone L., de Falco C., Computational Particle Mechanics, Springer. DOI: [10.1007/s40571-024-00864-2](https://doi.org/10.1007/s40571-024-00864-2), full text <https://rdcu.be/d20u2>
- 2024 **Review activity**, Reviewer for the Journal of Computational Particle Mechanics, Springer
- 2023 **Proceedings of Eccomas - Particles 2023**, *GPUs Based Material Point Method for Compressible Flows*, Baioni P.J., Benacchio T., Capone L., de Falco C. VIII International Conference on Particle-Based Methods, DOI: [10.23967/c.particles.2023.026](https://doi.org/10.23967/c.particles.2023.026)
- 2023 **Proceedings of SIMAI 2023**, *MS08 - Particle-Based Methods In Applied And Industrial Sciences*, Baioni P.J., Crescenzo N., Fois M., Moreno Martinez L. The XVI biennial congress of SIMAI, link: <https://web.unibas.it/simai2023/BookOfAbstracts.pdf>
- 2023 **Proceedings of SIMAI 2023**, *Material Point Method for Compressible CFD on GPUs*, Baioni P.J., Benacchio T., Capone L., de Falco C., Pellegrino E. The XVI biannual congress of SIMAI

FOSS Software

- 2020 Development of a C++14 code for a numerical methods and programming university course project, implementing a Deep Neural Network (DNN) from scratch, relying on the C++ Standard Library and Eigen; code available at <https://github.com/pjbaioni/neural-net>

Professional Memberships

- 2023 – present Member of INdAM, National Scientific Computing Group (GNCS)
- 2023 – present Member of Italian Society for Applied and Industrial Mathematics (SIMAI)
- 2022 – present Enrolled in the Order of Engineers of the Metropolitan City of Milan

Languages

- Italian Native speaker
- English C1 (TOEIC 2016); teaching MSc courses in English since 2022